

CONTACT INFORMATION	Department of Physics, University of Milano-Bicocca Piazza della Scienza, 3, Milan MI 20126, Italy	E-mail: weichen.wang@unimib.it https://weichenstars.github.io
EDUCATION	Johns Hopkins University, Baltimore MD, United States 9/2016 - 12/2022 Ph. D. in Astrophysics, Department of Physics and Astronomy Thesis Advisor: Susan Kassin (STScI); JHU Advisor: Timothy Heckman Thesis title: <i>Observational studies of the relation between galactic winds and star formation at half the age of the Universe</i> Tsinghua University, Beijing, China 8/2012 - 7/2016 B. Sc. in Physics, Department of Physics Thesis Advisor: Shude Mao	
RESEARCH POSITIONS	University of Milano-Bicocca, Milan MI, Italy 12/2022 - now Postdoc researcher, Department of Physics Advisor: Sebastiano Cantalupo University of California, Santa Cruz CA, United States 2020; 7-9/2015 Visiting student, Department of Astronomy Advisors: Sandra Faber, David Koo	
RESEARCH INTERESTS	Spectroscopic Surveys of Galaxies & Their Environment; Galaxy Formation at Cosmic Noon and Earlier Epochs; The Circumgalactic Medium	
PUBLICATION HIGHLIGHTS	Four first-author papers published in <i>Nature Astronomy</i>, <i>ApJ</i>, and <i>MNRAS</i>. A total of 17 publications with 1000+ citations in the past 3 years; h-index: 15 Click on this ADS link for full publication records. Highlight 1: Early-Epoch Galaxy Formation in Extreme Environments [5] W. Wang, S. Cantalupo, A. Pensabene et al. <i>Nature Astron.</i>, 9, 710 (2025): <i>A Giant Disk Galaxy Two Billion Years After The Big Bang</i> (Editorial Highlight) [4] A. Pensabene, S. Cantalupo, W. Wang et al. <i>A&A</i>, 701, A120 (2025): <i>ALMA survey of a massive node of the Cosmic Web at $z \sim 3$: II. A dynamically cold disk galaxy in the proximity of a hyperluminous quasar</i> Highlight 2: Relations between Galaxies and Gas Outflows [3] W. Wang, S. A. Kassin, S. M. Faber, D. C. Koo et al., <i>ApJ</i>, 930, 146 (2022): <i>The Baltimore Oriole's Nest: Cool Winds from the Inner and Outer Parts of a Star-Forming Galaxy at $z = 1.3$</i> Highlight 3: Galaxy Star Formation and Dust [2] W. Wang, S. A. Kassin, C. Pacifici et al., <i>ApJ</i>, 869, 161 (2018): <i>Galaxy Inclination and the IRX-β Relation: Effects on UV Star Formation Rate Measurements at Intermediate to High Redshifts</i> [1] W. Wang, S. M. Faber, F.-S. Liu et al., <i>MNRAS</i>, 469, 4063 (2017): <i>UVI colour gradients of $0.4 < z < 1.4$ star-forming main-sequence galaxies in CANDELS: dust extinction and star formation profiles</i>	

SURVEY SCIENCE AND TECHNICAL EXPERIENCE	JWST galaxy surveys: (1) Serving as the deputy lead for JWST program GO1835, a 20-hour spectroscopic survey of over 40 galaxies in two $z \sim 3$ QSO fields; responsible for the planning, implementation, and analysis of the NIRSpec and NIRCам observations; (2) Core team member to develop a novel spectroscopic mode for NIRSpec MSA “slitlet stepping”; co-leading the effort of observation planning tool development and implementation for a 74-hour JWST galaxy survey GARDEN (GO2123). Keck survey: Reducing, organizing, and analyzing the spectra of $\simeq 200$ $z=1$ galaxies as the member of a large Keck/DEIMOS survey HALO7D (P.I.: GuhaThakurta).
AWARDS	Giovani Talenti (Young Talents Award) by U. Milano-Bicocca & Accademia Nazionale dei Lincei of Italy, 2025 The International Astronomical Union Travel Grant, 2019 First-year student award, Johns Hopkins Physics & Astronomy Department, 2016 National Astronomical Observatory of China Scholarship, 2016 Award of excellence by Tsinghua University: 清华学堂计划, 2013-2016
MEDIA	Press releases by the European Space Agency, Nature Portfolio, U. Milano-Bicocca, Caltech, Swinburne U. (reporting W. Wang+ 2025): “<i>Big Wheel</i>”, a surprisingly large disk galaxy discovered in the early universe. News coverage by Scientific American, The Independent, Newsweek, Daily Mail, Il Giornale: <i>JWST Spots Giant Spiral Galaxy Shockingly Early in Cosmic History</i>.
TALKS	Invited Conference Talks: The Controversial Fate of High-z Galaxy Protoclusters, Spain, 2025 Galaxies and Diffuse Gas in Large-Scale Overdense Environments, Italy, 2024 Department Seminars: Tsinghua University/Peking University/NAOC, China, 2023/2024 University of Missouri, U.S., 2020 Recent Contributed Talks (Selected): The Baryon Cycle from Reionization to the Cosmic Noon, Chile, 2025 Tracing Cosmic Evolution with Galaxy Clusters, Italy, 2025 Observing and Simulating Galaxy Evolution in the Era of JWST, Switzerland, 2024 What Matter(s) Around Galaxies 2024, Italy, 2024
SERVICE	SOC/LOC member, Conference What Matter(s) Around Galaxies, 2024 Referee for The Astrophysical Journal, Astronomy and Astrophysics
OBSERVATION PROGRAMS (SELECTED)	Summary: 100+ hours on JWST and 120+ hours on VLT & ALMA as the core member. Programs joined as P.I. and core member are marked with [P] and [C]. [C] ALMA Cycle-12 Program (P.I.: A. Pensabene; 27 hours): <i>The Big Wheel Galaxy: a unique laboratory to study the physics of early disk formation in overdense environments</i>, 2025-2026 JWST Cycle-4 Program (P.I.: B. Wang; 20 hours): <i>Extremely massive galaxies in the early universe? Confirming the nature of the most</i>

model-breaking object by hunting for stellar absorption features, 2025-2026

[C] VLT P112 Program (P.I.: S. Cantalupo; 84 hours): *Connecting the dots with MUSE: the Cosmic Web in emission around a massive structure at $z=3$* , 2024-2025

JWST Cycle-2 Program (P.I.: Kassin; 68 hr): *Galaxy angular momentum alignment with filaments at $z\sim 3$: The effect of large scale structure on galaxies*, 2024

[C] JWST Cycle-1 Program (P.I.: S. Kassin; 74 hours): *A Pathfinder for JWST Spectroscopy: Deep High Spectral Resolution Maps of Galaxies over $1 < z < 6$* , 2023

[C] JWST Cycle-1 Program (joined in 2022 with major contributions; P.I.: S. Cantalupo; 24 hours): *Unraveling the Knots of Gaseous Cosmic Web Filaments at $z\sim 3$ through H-alpha Emission Observations*, 2023

[P] ALMA Cycle-7 Program (P.I.; 15 hours): *Does molecular gas follow the motion of ionized gas inside typical high-redshift star-forming galaxies?* Observations not completed due to weather and the impact of COVID-19 in Chile, 2021

MENTORSHIP

Xiaohan Wang, Visiting Ph.D student of Tsinghua University, 2025
JWST NIRSpec studies of galaxy metallicity in the high-redshift cosmic web
Resulting in one research paper completed & to be submitted to A&A.

M. Francesca Ubaldi, Bicocca undergraduate in physics major, 2024 (thesis project)
Relations between galaxy colors and morphology with JWST medium-band imaging
Now pursuing Masters in Astronomy degree in Bicocca.

Ying Qin, Johns Hopkins undergraduate in physics, 2021-2023
Studying the Mg II emission and ionizing photons from low-mass galaxies at $z \sim 1$
Resulting in two conference poster presentations, including one at the AAS Annual Meeting; Now pursuing a career as lawyer in Washington D.C.

TEACHING EXPERIENCE

Volunteer Teaching Assistant, Laboratory of Data Analysis in Astrophysics
University of Milano-Bicocca, Spring 2023

Teaching Assistant, General Physics I for Biological Science Majors
Johns Hopkins University, Fall 2016

Teaching Assistant, General Physics Laboratory
Johns Hopkins University, Fall 2016

OUTREACH ACTIVITIES

Science outreach day at University of Milano-Bicocca, 2024

A community event open to students from local primary/middle schools in Milan;
co-designing the exhibition about astrophysical science with JWST

Member of the Astro Scholars program, Johns Hopkins University, 2021-2022

An annual program about astrophysics and coding for college students from under-represented backgrounds; serving as a core member of the hiring & education team

Member of the Physics and Astronomy Graduate Students Outreach Team (PAGS),
Johns Hopkins University, 2017-2019

Supporting visits of students from Baltimore local primary/middle schools around once per semester and teaching fundamental physics with educational demos